

LOSO Experiment B_ypr

YPR only (both hands), no Nothing

Generated 2026-05-11 11:50:47

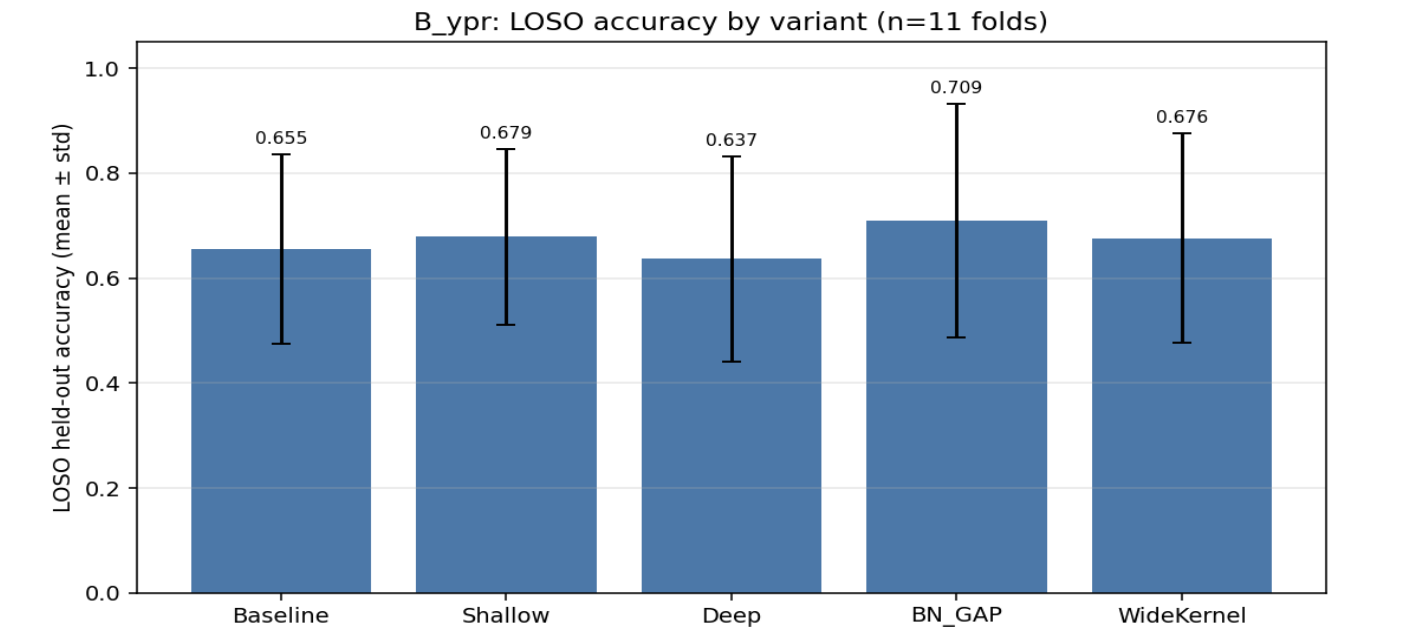
Experiment configuration

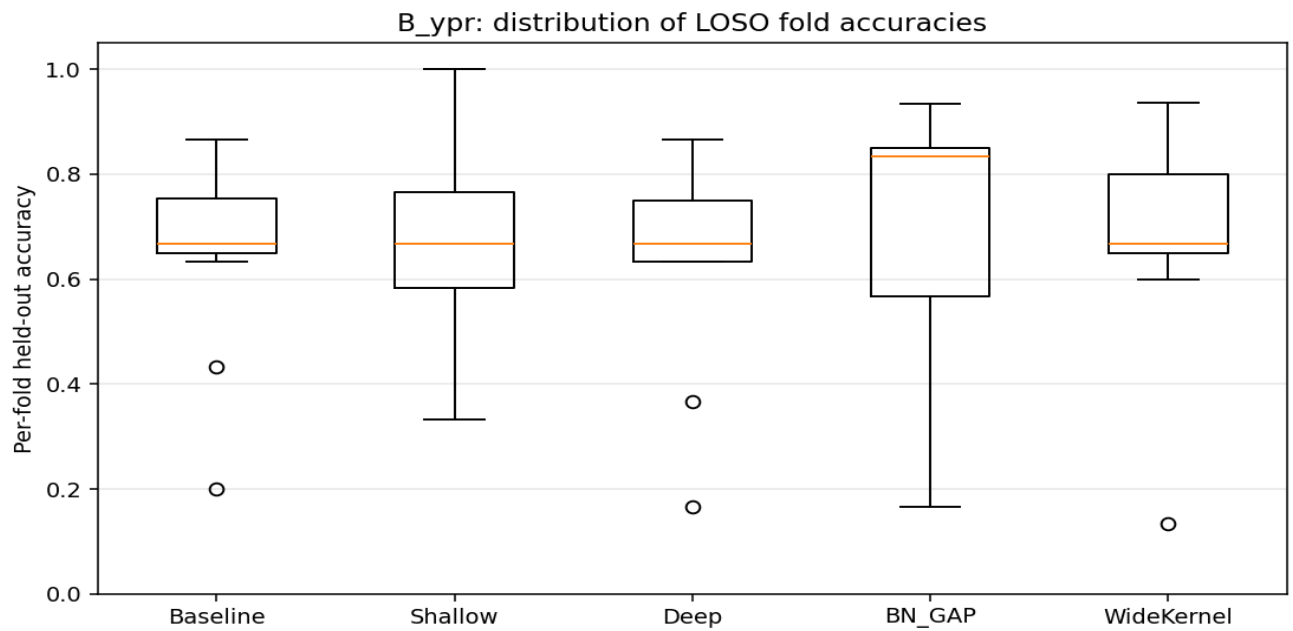
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	YPR
Hands	left, right
Sensor channels	78
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	11: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

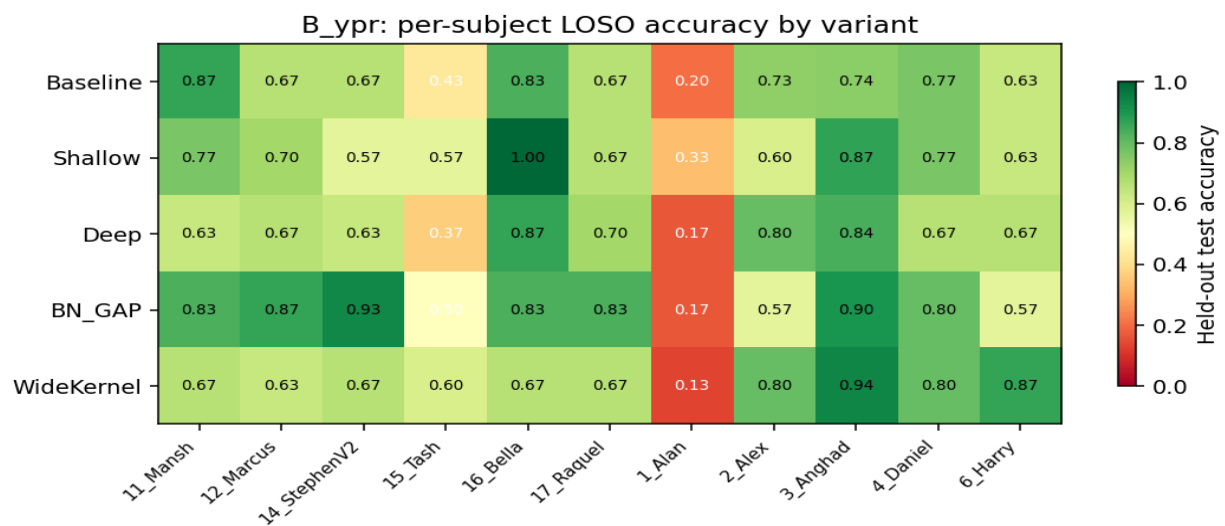
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
Baseline	100,646	0.6553	0.1809	0.2000	0.8667	0.8447
Shallow	56,774	0.6792	0.1676	0.3333	1.0000	0.8178
Deep	191,334	0.6369	0.1956	0.1667	0.8667	0.8188
BN_GAP	44,646	0.7094	0.2228	0.1667	0.9333	0.8758
WideKernel	106,534	0.6760	0.1993	0.1333	0.9355	0.7997

Best variant by LOSO mean: **BN_GAP**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	Baseline	Shallow	Deep	BN_GAP	WideKernel
11_Mansh	301	0.867	0.767	0.633	0.833	0.667
12_Marcus	301	0.667	0.700	0.667	0.867	0.633
14_StephenV2	301	0.667	0.567	0.633	0.933	0.667
15_Tash	301	0.433	0.567	0.367	0.500	0.600
16_Bella	301	0.833	1.000	0.867	0.833	0.667
17_Raquel	301	0.667	0.667	0.700	0.833	0.667
1_Alan	301	0.200	0.333	0.167	0.167	0.133
2_Alex	301	0.733	0.600	0.800	0.567	0.800
3_Anghad	300	0.742	0.871	0.839	0.903	0.935
4_Daniel	301	0.767	0.767	0.667	0.800	0.800
6_Harry	301	0.633	0.633	0.667	0.567	0.867

Variant: Baseline

Params: 100,646

LOSO mean accuracy: 0.6553 ± 0.1809 (min 0.2000, max 0.8667)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	1/5	5/5	5/5	5/5	5/5	5/5	0.867
12_Marcus	1/5	4/5	5/5	5/5	5/5	0/5	0.667
14_StephenV2	0/5	5/5	0/5	5/5	5/5	5/5	0.667
15_Tash	0/5	5/5	5/5	3/5	0/5	0/5	0.433
16_Bella	3/5	5/5	2/5	5/5	5/5	5/5	0.833
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
1_Alan	0/5	5/5	0/5	0/5	1/5	0/5	0.200
2_Alex	5/5	4/5	5/5	0/5	3/5	5/5	0.733
3_Anghad	5/5	5/5	5/5	1/5	6/6	1/5	0.742
4_Daniel	4/5	5/5	5/5	0/5	4/5	5/5	0.767
6_Harry	5/5	5/5	5/5	0/5	4/5	0/5	0.633
MEAN	0.527	0.964	0.764	0.436	0.764	0.473	0.655

Variant: Shallow

Params: 56,774

LOSO mean accuracy: 0.6792 ± 0.1676 (min 0.3333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	2/5	5/5	1/5	5/5	5/5	0.767
12_Marcus	5/5	5/5	5/5	0/5	5/5	1/5	0.700
14_StephenV2	0/5	5/5	1/5	1/5	5/5	5/5	0.567
15_Tash	2/5	5/5	5/5	4/5	1/5	0/5	0.567
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
1_Alan	2/5	3/5	5/5	0/5	0/5	0/5	0.333
2_Alex	0/5	4/5	4/5	0/5	5/5	5/5	0.600
3_Anghad	5/5	5/5	5/5	1/5	6/6	5/5	0.871
4_Daniel	3/5	5/5	5/5	0/5	5/5	5/5	0.767
6_Harry	3/5	5/5	5/5	0/5	1/5	5/5	0.633
MEAN	0.636	0.891	0.909	0.218	0.764	0.655	0.679

Variant: Deep

Params: 191,334

LOSO mean accuracy: 0.6369 ± 0.1956 (min 0.1667, max 0.8667)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	4/5	5/5	0/5	5/5	0/5	0.633
12_Marcus	0/5	5/5	5/5	5/5	5/5	0/5	0.667
14_StephenV2	5/5	5/5	4/5	0/5	5/5	0/5	0.633
15_Tash	1/5	5/5	5/5	0/5	0/5	0/5	0.367
16_Bella	4/5	5/5	5/5	2/5	5/5	5/5	0.867
17_Raquel	1/5	5/5	5/5	5/5	5/5	0/5	0.700
1_Alan	0/5	0/5	5/5	0/5	0/5	0/5	0.167
2_Alex	5/5	4/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	5/5	5/5	5/5	4/5	6/6	1/5	0.839
4_Daniel	0/5	5/5	5/5	0/5	5/5	5/5	0.667
6_Harry	5/5	4/5	2/5	0/5	4/5	5/5	0.667
MEAN	0.564	0.855	0.927	0.291	0.800	0.382	0.637

Variant: BN_GAP

Params: 44,646

LOSO mean accuracy: 0.7094 ± 0.2228 (min 0.1667, max 0.9333)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	5/5	2/5	4/5	4/5	0.833
12_Marcus	5/5	5/5	5/5	5/5	5/5	1/5	0.867
14_StephenV2	5/5	5/5	5/5	3/5	5/5	5/5	0.933
15_Tash	0/5	5/5	5/5	4/5	1/5	0/5	0.500
16_Bella	5/5	5/5	0/5	5/5	5/5	5/5	0.833
17_Raquel	5/5	5/5	5/5	4/5	5/5	1/5	0.833
1_Alan	0/5	5/5	0/5	0/5	0/5	0/5	0.167
2_Alex	0/5	4/5	5/5	2/5	1/5	5/5	0.567
3_Anghad	2/5	5/5	5/5	5/5	6/6	5/5	0.903
4_Daniel	4/5	5/5	5/5	0/5	5/5	5/5	0.800
6_Harry	5/5	2/5	5/5	0/5	0/5	5/5	0.567
MEAN	0.655	0.927	0.818	0.545	0.655	0.655	0.709

Variant: WideKernel

Params: 106,534

LOSO mean accuracy: 0.6760 ± 0.1993 (min 0.1333, max 0.9355)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	5/5	0/5	5/5	0/5	0.667
12_Marcus	0/5	4/5	5/5	5/5	5/5	0/5	0.633
14_StephenV2	0/5	5/5	4/5	1/5	5/5	5/5	0.667
15_Tash	0/5	5/5	5/5	4/5	4/5	0/5	0.600
16_Bella	5/5	5/5	0/5	0/5	5/5	5/5	0.667
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
1_Alan	4/5	0/5	0/5	0/5	0/5	0/5	0.133
2_Alex	0/5	4/5	5/5	5/5	5/5	5/5	0.800
3_Anghad	5/5	5/5	5/5	3/5	6/6	5/5	0.935
4_Daniel	2/5	5/5	5/5	2/5	5/5	5/5	0.800
6_Harry	5/5	5/5	5/5	5/5	1/5	5/5	0.867
MEAN	0.564	0.873	0.800	0.455	0.818	0.545	0.676